

LAW OF LARGE NUMBERS

=> Given a sequence of indep. rand. vars.

$$\bar{X}_1, \bar{X}_2, \dots, \bar{X}_n \quad \text{w/ } E(\bar{X}_i) = \mu, \text{Var}(\bar{X}_i) = \sigma^2$$

The sample mean $\bar{X} = \frac{1}{n} \sum_{i=1}^n \bar{X}_i$
will converge to μ as $n \rightarrow \infty$.

=> **Formally:** $P(|\bar{X}_n - \mu| > \epsilon) \rightarrow 0$ as $n \rightarrow \infty$

-> **PROOF:** Since $\bar{X}_n$ INDEPENDENT

$$\bullet E(\bar{X}_n) = \frac{1}{n} \sum_{i=1}^n E(\bar{X}_i) = \mu$$

$$\bullet \text{Var}(\bar{X}_n) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(\bar{X}_i) = \frac{\sigma^2}{n}$$

$$\text{as } n \rightarrow \infty, \text{Var}(\bar{X}_n) \rightarrow 0$$

$$P(|\bar{X}_n - \mu| \geq \epsilon) \leq \frac{\text{Var}(\bar{X}_n)}{\epsilon^2}$$

⚡
Chebyshev
Ineq.

$$= \frac{\sigma^2}{n \epsilon^2} \rightarrow 0$$

$$\text{as } n \rightarrow \infty$$

